

Technology Stack

Date September 2026
Team ID: 6abc95cbdfc814753aa5fe42
Project Name: EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks 2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, JavaScript, React and Vite	React provides reusable components for an interactive learning interface. Vite provides fast development and production builds.
2	Backend / Server-Side	Node.js with Express.js	Handles application requests and connects the frontend with the Gemini AI service. It also manages server-side application logic.
3	Database / Data Storage	Firebase Firestore	Stores user details, learning history, saved questions and generated learning content in a cloud database.
4	Cloud / Hosting / Deployment	Vercel	Hosts the EduGenie web application and provides a fast public URL for accessing the project.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
5	Version Control & CI/CD	Git and GitHub	Helps team members work together, manage different versions of the project, maintain source code and track changes.
6	Third-Party APIs / Other Tools	Google Gemini API, Google Fonts, Visual Studio Code, Node.js & npm, Chrome DevTools	Gemini API provides AI-powered answers, explanations, summaries and personalized learning support. VS Code is used for coding, npm manages packages, Google Fonts supports the interface, and Chrome DevTools is used for testing and debugging.